Paper 11 — Why Very Wide ReLU Networks Can Be Trained

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A Convergence Theory for Deep Learning via Over-Parameterization

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The paper in one minute

Question: How can ordinary GD and SGD fit training data with deep ReLU networks when their objective is non-convex *and* non-smooth?

Model/Method: Sufficiently wide networks initialized at random. Prove that near initialization, large loss forces a nonzero gradient and a semi-smoothness bound controls each update.

Finding: Under data-separation and width assumptions, GD and minibatch SGD reach arbitrarily small squared training loss in polynomially many iterations, with logarithmic dependence on accuracy. Certain CNNs and ResNets are covered too.

Why we care: Architecture and width can make the region actually visited by training benign; an optimizer need not explicitly search for Hessian negative-curvature directions.

1. Start with the difficulty, not the theorem

A deep neural network multiplies and composes weight matrices, and uses kinks like $\phi(z) = \max\{z, 0\}$. Its training objective is generally non-convex and not differentiable everywhere. *Having enough parameters to fit the data* does not by itself show that *GD starting randomly will find those parameters*.

The authors ask for both *efficient trainability* and a dependence on depth L that is polynomial rather than exponential. Their central idea is local: we need good optimization geometry *near the random initialization and training trajectory*, not across the entire parameter space.

2. The model and two essential conditions

For the fully connected main analysis, let $W_1, \dots, W_L$ be the trainable hidden-layer matrices, A the input map, and B the output map. The forward pass is

$$h_{i,0} = \phi(Ax_i), \qquad h_{i,\ell} = \phi(W_\ell h_{i,\ell-1}), \qquad \widehat{y}_i = Bh_{i,L}.$$

The square-loss training objective is $F(W) = \frac{1}{2}\sum_{i=1}^{n} \|\widehat{y}_i - y_i^*\|_2^2$. Two conditions matter:

- **Nondegenerate data:** distinct normalized training inputs have pairwise separation at least $\delta > 0$. Identical inputs with conflicting labels cannot both be fitted.
- **Overparameterization:** each hidden layer has sufficiently large width m, polynomial in the sample count n, depth L, and inverse separation $1/\delta$ (plus problem-specific dimensions/factors).

Do not overread "global minimum." In the square-loss result, $F(W) \le \varepsilon$ means the nonnegative training objective becomes arbitrarily small. It is not a guarantee of test accuracy, and it does not say that every stationary point everywhere is benign.

3. Worked example: why a large gradient can force the loss down

A teaching example of the local linearized prediction map, not the exact network or proof in the paper. Suppose two training residuals are $e = \widehat{y} - y^* \in \mathbb{R}^2$, and the Jacobian of the predictions with respect to three parameters is

$$J = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad K = JJ^\top = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Step 1 — The loss and gradient. With $F = \frac{1}{2}\|e\|_2^2$, the parameter gradient is $\nabla_\theta F = J^\top e$. Therefore,

$$\|\nabla_\theta F\|_2^2 = e^\top JJ^\top e = e^\top Ke.$$

Step 2 — Why distinct prediction directions matter. The eigenvalues of K are 1 and 3. Thus for every residual e,

$$\|\nabla_\theta F\|_2^2 = e^\top Ke \geq \|e\|_2^2 = 2F.$$

If $F > 0$, the gradient cannot be zero. This is the intuition behind a *gradient-dominance-like lower bound*.

Step 3 — A numerical GD update. In the exactly linearized model, a parameter step of size η updates residuals by $e^+ = (I - \eta K)e$. Take $\eta = \frac{1}{4}$ and $e = (1, -1)^\top$, an eigenvector with eigenvalue 1:

$$e^+ = \tfrac{3}{4}e, \qquad F_0 = \tfrac{1}{2}(1^2 + (-1)^2) = 1, \qquad F_1 = \left(\tfrac{3}{4}\right)^2 = 0.5625.$$

The residual is not trapped at a nonoptimal stationary point; the training loss decreases in this model.

What transfers to the paper: overparameterization and separated inputs are used to prevent the different sample gradients from canceling each other near initialization. The actual deep ReLU proof is significantly harder: the Jacobian changes with the weights and activation signs can flip.

4. Theorem 3: no bad critical points near initialization

The paper proves (with high probability and within its specified neighborhood of random initialization) a lower bound of the form

$$\|\nabla F(W)\|_F^2 \;\geq\; c\, F(W), \qquad c = \Omega\left(\frac{\delta m}{d\, n^2}\right) > 0,$$

where d denotes the output dimension in the paper's notation. There is also an upper gradient-norm bound. The authors call this a *no critical point* property: positive loss cannot coexist with zero gradient *in the controlled neighborhood*. It resembles a Polyak–Łojasiewicz inequality but is derived for the composite network locally, not assumed globally.

5. Why the ReLU kink creates a second obstacle

The gradient bound alone does not prove that GD decreases the loss at every step. The classic smooth descent lemma assumes a Lipschitz gradient, but ReLU changes slope abruptly at zero. For example, $\phi(-0.1) = 0$ and $\phi(0.2) = 0.2$; across zero the derivative jumps from 0 to 1. The paper therefore proves a different *semi-smoothness* inequality controlling the loss after a small weight perturbation.

6. Theorem 4: semi-smoothness replaces ordinary smoothness

Let Δ be a small update and W remain near initialization. The paper's bound has the schematic form

$$F(W+\Delta) \le F(W) + \langle \nabla F(W), \Delta \rangle + C_2\|\Delta\|^2 + C_1\|\Delta\|\sqrt{F(W)}.$$

The constants and precise matrix norms are in Theorem 4; this displays its structure. Unlike the textbook smoothness lemma, the bound contains an additional term $\|\Delta\|\sqrt{F}$ reflecting nonsmooth activation changes. With enough width and a suitably small step, the authors show these error terms can be controlled.

7. Combine the two results: why convergence becomes geometric

Step 1. A GD update $\Delta = -\eta\nabla F(W)$ makes the linear term $-\eta\|\nabla F(W)\|^2$ negative.
Step 2. The semi-smoothness result controls the terms that could otherwise spoil the loss decrease.
Step 3. The gradient lower bound replaces $\|\nabla F\|^2$ by a multiple of F; the proof obtains a contraction of the form

$$F(W_{t+1}) \le (1-\gamma)F(W_t), \quad 0<\gamma<1, \quad \Longrightarrow \quad F(W_T) \le (1-\gamma)^T F(W_0).$$

Step 4. Because the loss contracts fast enough, weights do not wander outside the initialization neighborhood where Theorems 3 and 4 apply. This closes the argument.

Why this is more than "wide networks have no saddles." The theorem controls a high-probability, *local* region and the particular GD/SGD trajectories. It does not characterize every critical point of the full non-convex landscape. The proof uses both geometry (no bad local critical points in the region) and dynamics (a guaranteed useful update).

8. Main formal guarantees: what is actually proved

Result	Meaning under the theorem's specified assumptions
Theorem 1	GD reduces squared training loss to ε in $T = \mathrm{poly}(n, L, 1/\delta)\log(1/\varepsilon)$ iterations, with sufficiently large m and small step size.
Theorem 2	Minibatch SGD also attains logarithmic dependence on $1/\varepsilon$; the stated iteration bound additionally depends on batch size b and logarithmic width factors.
Theorems 3–4	Local lower gradient bound (no positive-loss critical point) and semi-smoothness, the two technical ingredients behind convergence.
Theorems 5–7	Conditional extensions to other output losses, a specified CNN family, and a simple residual-network architecture.

A caveat about SGD: geometric convergence is not automatic for arbitrary stochastic objectives; here the training-data interpolation structure and problem assumptions are crucial.

9. How the proof works, without reproducing the appendices

1. **At random initialization:** concentration controls forward and backward signal propagation across L layers without allowing exponential blow-up in depth.
2. **After small perturbations:** show that only a controlled number of ReLU activation signs change and hidden representations remain close to their initialized values.
3. **Lower-bound gradients:** distinct data points give sufficiently separated hidden representations, preventing contributions from many examples from canceling completely.
4. **Control every update:** prove semi-smoothness and combine it with the gradient bound so GD/SGD decreases loss and remains in the trustworthy region.

10. What the figure and experiments support

Figure 1 shows empirical 2D loss slices around points along an SGD trajectory on CIFAR-10, using gradient and a negative-curvature direction after smoothing. It illustrates why following the negative gradient can still provide substantial loss decrease even though negative curvature exists. *This is an illustration consistent with the theory, not a proof that all practical training stays inside the theorem's neighborhood.* The authors explicitly acknowledge practical weight movement can exceed their bounds.

11. Important assumptions and limitations

- **Nondegenerate data and large width:** the main results require sample separation δ and potentially very conservative polynomial width requirements.
- **Specific initialization and training setup:** Gaussian initialization, carefully tuned learning rates, and staying close to initialization; the main formulation holds A, B fixed while training hidden matrices W_ℓ.
- **Training not generalization:** arbitrarily small squared training error or perfect *training* classification accuracy does not imply low test error.
- **Not every arbitrary loss has the same guarantee:** the nonconvex output-loss extension distinguishes gradient-dominant, convex, and general nonconvex cases.
- **Architecture claims have scope:** CNN and ResNet extensions cover particular specified constructions, not every modern architecture.

12. Connections to our survey

Paper	How it differs from Allen-Zhu et al.
Lee (2016)	General asymptotic avoidance of strict saddles under smoothness; no global training-fit guarantee.
Jin (2017)	Perturbations yield approximate second-order stationary points in finite time; here local geometry can yield *global training fit* without explicit escape.
Du (2019)	Another global-fit theory for wide deep nets; emphasizes smooth activations and GD, while Allen-Zhu treats ReLU, SGD, and polynomial depth dependence for fully connected nets.
Wu (2025)	Analyzes nonsmooth stationary points and escape neurons in shallow ReLU-like nets rather than wide-network training convergence.

13. Quick reference and Papers Master note

Term	Meaning
Overparameterized	m large enough (polynomial in $n, L, 1/\delta$, with other factors) for local high-probability control.
No-critical-point bound	$\|\nabla F\|^2 \geq cF$: positive training loss forces a nonzero parameter gradient.
Semi-smoothness	Descent bound with $\|\Delta\|^2$ and $\|\Delta\|\sqrt{F}$ correction terms.
Geometric rate	$F_T \leq (1-\gamma)^T F_0$, hence $T = O(\log(1/\varepsilon))$ for fixed problem parameters.

Our notes (copy into Papers Master): Wide randomly initialized deep ReLU networks can be trained by GD/SGD to arbitrarily small squared training loss in polynomial time under data-separation and width assumptions. The proof combines a local gradient lower bound (no positive-loss critical points) and semi-smoothness, keeping iterates near initialization. Results concern training fit, not generalization; width bounds can be conservative.

Primary source: Allen-Zhu, Li and Song, *ICML* (2019), Theorems 1–7 and Sections 3–6. `https://proceedings.mlr.press/v97/allen-zhu19a/allen-zhu19a.pdf`. Worked Jacobian example is pedagogical, not from the original paper.